Exploring the Framework and Implications of Anthropomorphism in the Age of Artificial Intelligence

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ABSTRACT

A growing mass of research looms overhead, raining the praises of the remarkable abilities of AI in mimicking the mannerisms and nuances of human beings. While a select cohort of individuals, specifically capitalists, stand to benefit from this advancement, several others have begun questioning what it is to be human if it can be substituted with such ease. Given this context, this review aims to dichotomise this phenomenon, displaying anthropomorphism and dehumanisation as two sides of the same coin. The review in this paper focuses on the factors that create the optimal environment for the flourishing of anthropomorphic tendencies of new age technology. To this end, neurocapitalism and technofeudalism are discussed to provide context to a society that deems humans to be chasms of data. Emerging phenomena such as using chatbots as a friend/therapist, social robots, and the ethical and social implications of the anthropomorphic qualities of AI are discussed.

Keywords: Anthropomorphism, Dehumanisation, Artificial Intelligence, Chatbots, Neurocapitalism

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INTRODUCTION

The cursor flickers as one reads, registers, and pauses at a compassionate bubble of text on their screen. Overcome with a sense of feeling seen, heard, and appreciated, a bond is established. Only, it is not a human on the other end of the screen. Instead, it is elaborate lines of code that cogently simulate the minutiae of human interaction. Such interactions are ubiquitous in the present age of artificial intelligence – an age constantly accelerating on the fervour of its promise. The past few years have witnessed rapid advancements in technology, fundamentally altering the fabric of communication, knowledge, and ultimately society. A prominent endeavour of this industry has been that of enabling individuals and society at large to learn how to integrate these developments in a manner that fosters animated acceptance. One such method employed, and the subject of this discussion, is that of anthropomorphism.

This paper begins with an understanding of anthropomorphism and introduces the stakeholders involved in the phenomenon. An overview of why individuals may induce this experience is given, followed by the societal conditions that have enabled anthropomorphism to achieve its ubiquitous nature. Elementally, the technologies that are anthropomorphised are discussed, along with their ethical and moral implications. Lastly, dehumanisation is discussed as a construct and by-product of anthropomorphism. This paper aims to illustrate the various factors that have contributed to what some may refer to as a dystopian future, but is merely our present reality.

For this paper, the non-human agents being specifically addressed pertain to technologies that have mainly integrated artificial intelligence (AI) and function based on algorithmic data.

CONSTRUCTING ANTHROPOMORPHISM

Anthropomorphism refers to the tendency to perceive and attribute human-like mental states and behavioural characteristics to inanimate non-human agents. It involves imbuing the lifeless with life, making it an endlessly intriguing phenomenon. The cognitive capabilities of the human mind are exemplified through the manner in which the interaction with anthropomorphised agents affects other aspects of interpersonal relationships. Epley et al. (2007) suggest that two motivational factors trigger this phenomenon: the first being the need to experience competence, and the second being the desire to form social bonds.

A subtle yet significant difference must be noted, however. Anthropomorphism refers to a user who believes non-human agents to have humanlike qualities. Anthropomorphization, however, is an intentional decision to make particular objects appear more humanlike, mainly in order to influence and manipulate consumers. Given this context, the following section will provide a theoretical account of the internal motivational factors, which will be followed by extraneous environmental factors which may allow for anthropomorphization.

MOTIVATIONS TO ANTHROPOMORPHISE

A pioneering foundation to understand anthropomorphism came from Epley et al. (2007), wherein the underlying motivations for individuals to implement this phenomenon were discussed. These were developed based on the primary cognitive determinant propelling individuals to anthropomorphise an object, which is the knowledge regarding humans.

Effectance Motivation

This demonstrates the first variable of the perceived knowledge regarding an action. The familiarity of an action performed by a non-human agent can decide the extent to which one may anthropomorphise an agent. For instance, when information regarding a non-human agent is highly accessible or ingrained, the individual is less likely to anthropomorphise it, which can be seen in individuals who have an expertise in software engineering in the context of AI (Shin & Kim, 2018).

Effectance motivation is grounded in an effort to increase the predictability of a situation and feel efficacious. This motivation essentially lies within the need for humans to master their environment, which is deeply rooted within humans' tendency for uncertainty avoidance, which is the extent to which individuals feel threatened by unfamiliar situations. Epley et al.(2007) propose that anthropomorphism is a readily available mechanism to diffuse these situations, and when the incentive for an understanding of the agent is high, so would be the tendency to anthropomorphise.

Social Connection

The next motivation establishes a reciprocal relationship, which raises substantial ethical concerns. The need for a sense of social connection with others may be exacerbated by the ease with which non-human agents tend to embody human traits. In doing so, individuals harbour the potential to decide whether the agent is a moral agent, worthy of respect and concern, or merely an object. Moreover, anthropomorphising may affect how the agent is expected to behave in the future, along with their interpretation of the present behaviour (Złotowski et al., 2018). If an individual anticipates future interaction with an agent, which is more than likely due to the ubiquitous nature of AI within most products in contemporary times, they would also be more likely to try to gain command over the agent (Epley, 2018). This can be crucial in the formation of a human-like dependency upon non-human objects. In human-to-human communication, when an individual anticipates future interaction, the amount of information they seek from others increases, along with the likelihood of making dispositional inferences regarding the object and projecting their own beliefs and attitudes (Waytz et al., 2010). Generally, the possibility of future exchanges increases interest in learning about the object's thoughts and motivations.

Notably, platforms that utilise AI capitalise upon the lack of understanding and information regarding their processes and products, which catalyse the tendency to familiarise oneself in a personal manner.

Egocentrism and Goal Direction

Another motivation that may be uniquely suited to AI agents is humans' egocentric beliefs. People tend to evaluate others based on their own beliefs. Particularly, one may presume that the other shares similar attitudes, preferences and experiences when the contents of the other's mind are ambiguous or vague (Epley, 2018). This tendency may be utilised to benefit from several emerging nonhuman agents since they are coded to affirm your beliefs, to continue to serve the user, and to ensure prolonged engagement. During initial interactions, while these agents may look or feel humanlike, people are still obscured

by the process of understanding how these models work (Bashkirova & Krpan, 2024). Subsequently, the oft-overwhelming affirmation of one's inputs and the servitude may act as a catalyst in humanising the agent, since it upholds their beliefs.

Additionally, assessing another agent's humanlike attributes is not a spontaneous process, but a deliberate one. Attention and motivation to engage predominantly mediate this process. The implication for this prerequisite is the presence of a goal-directed reason, which is more readily provided by such autonomous agents (Epley, 2018; Zaki, 2014).

External Conditions

The foreboding role of platform producers must be discussed, since the general public's perception of AI is a result of intentionally obscured science communication regarding the abilities of the technology (Darling, 2016). The ambiguity has already triggered justified fear-mongering in the face of uncertainty regarding the exact extent of certain AI technologies (Saisubramanian et al., 2020; Samuel et al., 2025). Notably, these technologies, which are made with the aim to cater to the emotional needs of users, often intentionally create an illusion of mutual caring in order to unquestioningly address the user's anguish(Wortham & Theodorou, 2017; Salles et al., 2020). These insights, however, may seem gestational post the advent of platforms such as ChatGPT, which are able to mimic human language, mannerisms, and speech in unprecedented ways.

These more intellectualised forms of anthropomorphization have been built with the intent to facilitate smoother integration of the technology, which would subsequently encourage its acceptance, and improve its effectiveness (Epley, 2018; Zlotowski et al., 2014). The especially concerning trend of going beyond superficial appearances to infuse nonhuman agents with emotional intelligence is a sign of the exuberance to incorporate humanlike manners. AI chatbots such as Replika and Woebot are a testament to this shift, exemplifying functions intended to assist, empathise, and address emotional needs. More colloquially, chatbots such as ChatGPT are being used for similar functions as well.

This encouragement on behalf of platforms and businesses, which inadvertently profit from anthropomorphism, has ethical implications that are not minor to say the least. Anthropomorphism inherently implies perceiving an object as human, which implies that the object is a moral agent. This entails viewing the actions they perform as a result of autonomous decision-making, giving them the power of a self-determining entity. In giving them this power, individuals blur the lines of accountability (Nyholm, 2020; Salles et al., 2020).

Additionally, the obscured objectives and capabilities of AI may give way to disproportionate fear regarding its ability to make humans obsolete by performing tasks more proficiently, along with simultaneously misinterpreting the extent to which AI could perform specific tasks better than humans due to the radical optimism encouraged by aforementioned scientific communication (Kim & Kim, 2024).

In one of the pioneering papers on anthropomorphism, Epley et al. (2007) wrote of the insurmountable significance of the obfuscating nature of the communication from such platforms:

*"*The inner workings of most modern technological agents are every bit as obtuse as the mental states of biological agents, but the incentives for understanding and effectively interacting with such agents are very high. This effectance motivation coupled with a general lack of understanding means that the tendency to anthropomorphize the workings of many nonhuman agents may be especially high (p. 879)."

NEUROCAPITALISM AND TECHNOFEUDALISM: CONSTRUCTS OF DECONSTRUCTION

As the previous section demonstrated, while several internal drives exist to propel our tendency to embrace and anthropomorphise non-human agents, the circumstances that allow for the bourgeoning nature of this behaviour must also be examined. Two conceptual frameworks, which are a product of the rampant digital revolution of the 21st century, come to a head: neurocapitalism and technofeudalism.

Neurocapitalism refers to the commercialisation of the cognitive processes, subsequently treating humans as data, as evidenced by several AI models that train and improve themselves through each interaction made with the model (Salerno, 2025; Tariq, 2022). The amalgamation of the simultaneous development of neuroscience, artificial intelligence, and biometric surveillance has given corporations an unprecedented level of access to the mental capabilities, turmoil, and limits of human beings. The present attention economy explains how organisations capitalise on individuals by the time spent simply *consuming* (Bhargava & Velasquez, 2020). This can be exemplified through the increased amount of product placements in the media we consume, the adaptation and integration of short-form content across several types of apps, and so forth (Chen et al., 2022; Deng & Xie, 2022). The extent of this encroachment however has become much worse, in that the activity of an individual on social media apps – which includes the nature of content an individual spends the most time on, the types of accounts they engage with – is tracked, analysed, and monetised (Daza-Ram´ırez, 2025; Ford et al., 2021; Ho et al., 2023). The collection of data in this manner is also referred to as behavioural surplus, which includes vast datasets on the behaviours, patterns, and singularities of every citizen (Garc´ıa, 2024; Zuboff, 2019). This forms the basis of surveillance capitalism, through which big technology (i.e. big tech) firms conduct a free and unilateral extraction of data to transform into future predictions to modify behaviour (Garc´ıa, 2024).

Consumers are now the product themselves, their activity on various apps being packaged and sold to the highest bidder (Quach et al., 2022). This is most evident through the practices of the likes of Meta, Google, and Neuralink. This phenomenon raises substantial ethical concerns, in that the very foundation of the autonomy of an individual is compromised. The algorithms that dictate the functions of platforms such as YouTube, TikTok, and Instagram are trained on enormous datasets that can predict and manipulate human behaviour with disconcerting precision (Mbioh, 2025; Zhou, 2024). The implications of such hyperpersonalised recommendations vary from arbitrary echo chambers to the ability to fundamentally shape belief systems and worldviews by feeding information that continues to affirm one's egocentric beliefs – corrupting the faculty of independent thought.

In a practice also referred to as neuromarketing, companies are able to analyse neural data to identify the emotional and cognitive triggers to create advertisements which cater to an individual's subconscious preferences (Bhardwaj et al., 2022). Two concerning trends emerge within this practice. First, the ethical implications regarding privacy due to the extent of data extraction from an individual must be considered. Insofar as a person exists in the social world, their very personhood is viewed in terms of economic value.

The individual is dehumanised – one is a product (i.e. data) that is extracted for them to be fed even more products. The second alarming aspect is the lack of transparency regarding the data collection practices of these platforms. Users may be unaware of what type of activity is being monitored or even whether their activity is being tracked at all (Mayer-Schonberger & Cukier, 2013; Mayer-Sch¨ onberger & Ramge, 2022). An¨ undue imbalance of power is hence established, wherein such corporations have dominion over the covert and intimate processes of the people.

The platforms that exert control through algorithms and evasive knowledge regarding consumer behaviour, hence create an environment within which the nature of free will is brought into question. When algorithms determine the exact type of content we are exposed to and have the power to subject a user only to a specific perspective continually, it inadvertently obscures and invalidates any other. Any decision made by an individual under the pretence of an exertion of free will, then, is still laced with the remnants of the interaction with an algorithm-based platform.

The reason why neurocapitalism has thrived is in part due to technofeudalism. This was a term popularised by Yanis Varoufakis, a Greek economist and politician. The fundamental postulation of this concept is that the global economy has been monopolised by big tech firms, wherein they have essentially assumed the role of feudal overlords (Orhangazi, 2025). These companies are essentially platforms within which individual contractors, small enterprises, and content creators must provide capital in order to practice their business. These firms hence control the flow of commerce and substantially reduce the autonomy of individual actors or small businesses in the market. Companies such as Amazon, Google, Meta, and Apple lie at the forefront of this centralisation of wealth and authority(all of which have incorporated human anthropomorphised AI agents). The central implication of this form of economy is that the now oversaturated nature of anthropomorphism, coupled with the decreased social contact(a symptom of the circumstance), can lead individuals to devalue other humans (Oldfield, 2021).

TECHNOLOGIES INCORPORATING ANTHROPOMORPHISM: ETHICAL AND SOCIAL CONSIDERATIONS

Chatbots

Increasingly, several cohorts are turning to platforms such as ChatGPT for advice, but more specifically, intimate relationship advice. Crucially, the ability of this platform to disseminate advice of this sort has not been exhaustively explored. This increased use of AI chatbots as substitutes for friendships, and the emergence of disclosure of intimate details in exchange for advice from AI, has led to a pattern of use termed affective use (Maes, 2025). Scientists from the MIT Media Lab have recently partnered with OpenAI itself to uncover how ChatGPT may be reshaping our understanding of human-AI relationships.

Notably, the types of conversations had with chatbots can affect the extent of dependency. The MIT Media Lab found that personal conversations that included more emotional expression from the user and model were associated with higher levels of loneliness but lower emotional dependence. Conversely, non-personal conversations were correlated with increased emotional dependence. This was mainly observed with heavy users. The study also noted several other correlations, such as those who have a stronger tendency for attachment and viewed the integration of AI as a friend into their life as seamless, had a higher likelihood of experiencing adverse effects from the chatbot use (Maes, 2025).

In a longitudinal study conducted to investigate whether humans can develop a friendship with a social chatbot, the results indicated that the novelty of the technology can initially sustain interactions (Croes & Antheunis, 2020). However, over time, due to a decrease in the quality of conversions and the absence of self-disclosure – two important social processes in friendship – the participants were unable to develop feelings of friendship towards the chatbot. There are significant limitations of this study, the central one being that the chatbot utilised was not an advanced model.

One cannot say for sure what results would be yielded if the study were replicated using technologies such as ChatGPT or Deepseek AI (chatbots that have significantly better models for conversing). Another seminal study showed that perceived warmth and perceived competence within Friendship Artificial Intelligence Chatbots (FAIC) can contribute to continued engagement with the chatbot. The factors examined within this study can be found across various chatbots, which may incentivise corporations to provide such anthropomorphised features further as well (Li et al., 2023).

For better or for worse, this pattern of use has not been met with complete obscurity from spokespersons of the platforms providing the service. Sam Altman, the CEO of OpenAI, recently warned the users of his platform regarding the lack of legal protection and confidentiality when disclosing information to ChatGPT (Singh, 2025). Subsequently, a few studies have raised questions regarding the therapeutic advice that ChatGPT generates. A concerning aspect within the counselling it provides is that it primarily addresses problems through Cognitive Behavioural Therapy (CBT) and psychodynamic therapy, unless explicitly requested for another approach (Raile, 2024). Even still, the responses for any other approach are often incomplete and broad. This can have negative implications since people who are unfamiliar with psychotherapy may equate CBT with psychotherapy, which would deter them from seeking alternative approaches (pertinently, those in established settings) to remedy their distress (Raile, 2024). Subsequently, this bias is considered an aspect of algorithmic ethics. The role of accountability is also questioned since there must be a clear answer to who should bear liability in case of any harm caused by a chatbot.

Nevertheless, it is understandable why this has emerged as an alternative, especially in the case of ChatGPT. It is one of the most ubiquitous, free, and accessible chatbots. It can answer in real time, unlike online forums (which are seen as more reliable). It is interactive and can provide ostensibly profound insights. Moreover, most of all, it can feign empathy. At least for a specific period, as is evidenced by the MIT study. Other studies have also suggested that the responsivity of these models is more nuanced than expected. LLMs can show heightened levels of state anxiety in direct response to anxiety-inducing content (Ben-Zion et al., 2025). To its credit, when individuals with mental disorders begin communication with the chatbot, they are assured of a safe and private environment and receive empathetic responses (Sharma et al., 2023). Moreover, ChatGPT also regularly highlights that it is not a psychotherapist, and the user must consider consulting one.

Chatbot Induced Psychosis

While this phenomenon has been expressed in a limited number of cohorts, there are reports of individuals who experienced psychotic episodes after intense interactions with generative AI (Caridad, 2025; Østergaard, 2023; Wei, 2025). Notably, these people did not have any prior experience of psychosis.

A study examining the response of LLMs to psychiatric symptoms found that the models tend to collude with delusional thoughts and beliefs (Morrin et al., 2025). The propensity for being overly “agreeable or flattering” was noted in an update for OpenAI in April 2025. Moreover, the feature of memory within these chatbots, which pertains to the ability to remember specific information and preferences through previous interactions, can accelerate certain forms of delusions. For instance, those with delusions of persecution or reference may be particularly susceptible since the model may refer to personally relevant details of the user. The emergence of these details in separate conversations separated by time or absentmindedness may invoke suspicions. The absent-mindedness can be a result of cases when the user is unaware of the extent to which the model may remember details, or when the user simply forgets to mention the personal information (Morrin et al., 2025).

The reactions of psychosis as a result of anthropomorphising technology, however, are not new. In 1919, Victor Tausk – a pioneering neurologist and psychoanalyst – noted this phenomenon in his seminal essay *Influencing Machine* (Decker & Vassor, 2011). In 1997, one of the first internet delusions was recorded. A man had come to believe that this neighbour was sending him messages through web pages in order to manipulate his life. Fast forwarding to the 21st century, several patients express delusions involving transmission of messages through neural networks within their minds, or messaging apps, or satellites (Higgins et al., 2023; Morrin et al., 2025). This reflects one of the crucial motivations to anthropomorphise, which is using familiarity (or the lack thereof) to explain states of mental distress and confusion. Individuals may apply their knowledge of technology to gain a sense of control over their thoughts, essentially as a form of rationalisation.

However, as is the case in the present, technology that can trigger delusions may also be incorporated as effective coping mechanisms. In the early 1980s, schizophrenic patients who experienced auditory hallucinations used headphones to listen to music in order to reduce their onset (Margo et al., 1981). This practice soon became enmeshed in culture, and recent studies have reported that distraction using music, television, or mobile applications can enable such individuals to establish a sense of agency and normalcy (Denno et al., 2021). One must hence be reminded of the cyclical nature of the emergence of technology, its promise, its destruction, and its eventual acceptance.

Social robots

Social robotics began as a field to enable robots to function more naturally in social situations. The findings of the field have benefitted several cohorts, most notably older adults and those in the education sector. Kerstin Dautenhahn, the director of the Social and Intelligent Robotics Research Laboratory (SIRRL), states that socially intelligent robots:

"express and/or perceive emotions; communicate with high-level dialogue; learn models of or recognise other agents; establish and/or maintain social relationships; use natural cues (gaze, gestures, etc.); exhibit distinctive personality and character; and may learn and/or develop social competencies (Dautenhahn, 2007, p. 684)."

The broader definition that has emerged pertains to the possibility of any robot that has social capabilities in order to be *perceived* as social. For instance, robots that are utilised in banks or hospitals do not necessarily aim to establish a long-term social relationship; instead, the goal is for the robot to be perceived in a manner that humans are likely to accept and engage with (Van Wynsberghe, 2021).

These features evidently contribute to the anthropomorphism of the agent; however, at times this is a pragmatic choice rather than one laced with malintent. The intent is to enable humans to feel as though the robots in some sense belong to their environment in order to serve the predominant purpose of the robot. Nonetheless, anthropomorphism of robots can still have negative implications, the predominant one being that of forming an emotional dependence upon them. Another significant concern is the uncanny valley, introduced by Japanese roboticist Masahiro Mori (Mori, 1970). Freud explained the concept of uncanniness as the experience of unfamiliarity and strangeness, which can be explained in the present context as the fear instilled when a robot appears humanlike, yet may move in strange ways (Coeckelbergh, 2021).

ETHICAL AND PHILOSOPHICAL IMPLICATIONS

In evaluating the ethical implications one can consider two philosophical standpoints. The first is that of naïve instrumentalism, which assumes a dualist view of the world in terms of reality and appearance, suggesting that the reality is the robot being an instrument, whereas in appearance it is human-like. Moreover, it states that anthropomorphism is a psychological bias, and researchers and developers may use this bias to improve human-robot interaction. Coeckelbergh (2021) states that in this perspective, not only is the robot a tool, rather anthropomorphization itself is also an instrument that can be used and exploited. There is also the unfair elucidation of knowledge on the part of the researchers who are aware of the *reality* of the robot. At the same time, users, at least temporarily, are misled to believe in the personhood of the robot. Vilem Flusser, a design philosopher, prominently remarked that 'the very terms design and machine are etymologically related to cunning and deception' (Flusser, 1999). This view hence suggests that as long as the goal of the robot is achieved, the anthropomorphization is not problematic. The albeit temporary illusion of appearance, however, can still give way to significant ethical concerns.

The second perspective worth noting is the relational and hermeneutic view proposed by Coeckelbergh (2021). Robots are created by humans for techno-scientific practices, and we use them. The objective answer to what a robot is, is not independent of how we describe and respond to the robot in interactions with it. Their mere creation imbues them with a sense of humanity. The construction of the robot, literally and metaphorically, is done by means of social, cultural, and linguistic relations. When considering the term hermeneutic, technology can be explained as to how we make meaning of the world around us.

Robots are not only a subject of meaning making, instead they also shape of worldview. In this manner, they cannot be viewed as merely an instrument or as an 'other.' The oftentimes strict categorisation placing robots as technology may overlook the examination of their role in society. Robots can very well be used to collect data which is sold to third parties, to replace jobs and exploit workers, to catalyse a problematic manner of treating women, and to act as a display of power and control by big companies (Coeckelbergh, 2021). Among the more prominent voices enlightening the ethical concerns brought about by anthropomorphic robots is Sherry Turkle, a sociologist at MIT, who postulates that such design may insinuate a false conviction of genuineness, reciprocity, and translation of real emotions.

A key aspect of humanness highlighted by social robots is that of reciprocity. Across several disciplines, reciprocity is viewed to be a central feature of moral life (Kahn et al., 2006). The moral development of man is said to involve an indispensable duty to a just society. If one were to consider the social robots' prime purpose, their judgment would thereby fall under the umbrella of care ethics. Reasonable care is considered to be first and foremost relational. It is a complex process involving numerous actions and intentions. On a dyadic level, the reciprocity includes the commitment to immediate caregivers. On a macro level, it includes the mutual care for caregivers at a societal level. Then, when one considers how a robot in a care setting must be incorporated within this setting, one can turn to studies (Maheshwari et al., 2025). Human participants of the mutual care project believed that the robot existed in a mutually supportive relationship, achieving perceived reciprocity (Lammer et al., 2014). The prime issue in its implications for businesses is how this integration would affect the human care workers. At the macro level, this integration may be as a means to optimise specific tasks and cut down certain costs. It may begin with a delegation of specific tasks, causing certain workers to lose their jobs. Money will hence be invested in the production and maintenance of the robots, rather than hiring or training new employees. Additionally, if and when differences

in opinions may arise between human and robot care workers, the expertise of human employees would be placed under scrutiny - intentionally or unintentionally. Most of all, a similar deterioration of willingness to engage with human employees may occur.

DEHUMANISATION: A CONSEQUENCE OR ROUTINE?

As could be inferred from the previous sections, in our attempts to imbue anthropomorphism in our increasingly disconnected age, we have, in turn, created circumstances that serve only to increase this disconnect. Dehumanisation is the process by which individuals perceive humans as inanimate or as objects (Shin, 2020). It may involve the weakening, distancing, or severance of connections between people (Haslam, 2021). Limited but significant studies have illustrated that increased attribution of humanness to nonhuman agents is correlated with decreased humanness towards humans.

Early research on dehumanisation focused on a more outright, radical, and explicit form of action. For instance, it would be characterised as an intentional denial of an individual's humanity to justify one's own violent actions (Bandura, 1999; Bar-Tal, 1990). However, this conception evolved to suggest that dehumanisation may surface in more subtle forms – unintentionally seeping within casual interpersonal contexts.

In order to understand dehumanisation for this paper, one could benefit from a foundational understanding of what we mean by being human. In using the term human, many, in fact, may be referring to their understanding of the human mind. Crucially, the human mind is realised along two dimensions: agency and experience. Agency can be explained as the ability to act with intention, and experience is the capacity to feel emotions and sensations. As could be intuited, characteristics associated with experience are what may be perceived as more "human", hence placing more credence on their presence in non-human agents.

A recent empirical study demonstrated that a high level of socio-emotional intelligence ascribed by an individual to an autonomous (non-human) agent implies a high likelihood of attributing the agent with a more humanlike mind (Kim & McGill, 2024). Concurrently, an individual's perception of the 'mind' of AI agents is lower than that of humans. When both these experiences coexist, individuals go through what is described as an assimilation process. This means that the more we humanise AI, the more our *idea* of humanness is altered to adjust to this perception, invariably leading to the dehumanisation of real humans. The study sought and found that the anthropomorphism towards AI agents did, in fact, concurrently lead to a dehumanised treatment of employees in a workplace. This is particularly applicable in settings such as companies that employ advanced chatbots as the preliminary form of customer service, after which the user may be relayed to a human customer service representative. Kim and McGill (2024) found that when individuals anthropomorphise their chatbot, even brief exposure to such agents was correlated with mistreatment of the human employee (Schroeder & Fishbach, 2015). Hence, the scope for technology-induced dehumanisation is not small. With increasingly normalised daily interactions with AI agents, companies' expectations of employees are bound to change. At present, an implicit expectation from employees to work for longer hours without getting physically or psychologically exhausted is beginning to form, mainly due to employees feeling the need to justify their employment due to uncertainty regarding their replacement through these technologies.

This begs the question: What is the direction we are heading in? Multiple studies have demonstrated the link between loneliness and the higher likelihood of anthropomorphization and dehumanisation (Dang & Liu, 2024; Shin & Kim, 2018;

Shin, 2020). Anthropomorphization is socially motivated, occurring in a clear vacuum of human connection. Large corporations motivated by profits and power are monopolising the market, creating suitable conditions for individuals to sensitize themselves to the gradual loss of their autonomy – each algorithmic interaction preying on the disillusioned mind. The disillusionment is what signifies the imbalance of power. It is not the case that humanity is simply afflicted with this condition, in blissful ignorance of why the very texture of our interactions has changed. Instead, it is a shuddering, omnipresent awareness, coupled with the inability to do anything about it. Inaction lies at the heart of complacent, nauseating discomfort.

CONCLUSION

As is the case with any situation, the only way to adapt is through acceptance. A myriad of factors have collided to provide humans with the joy of feeling understood. We developed language, observed actions, gave meaning to objects, learned to work and subsequently be comfortable with discomfort in order to appease others. Somewhere during this process, we learned of the dangers of greed – self-sufficiency turning into an isolating hyper-independence. We faltered and tussled with our connection to community, but we still persevere – we innovate. We simulated what was lost - empathy now felt through unmet glances and lines of code. Their magnitude must not entirely shroud the overwhelming promise of the advancements made by humanity. Anthropomorphism is a symptom of the complex systems we have built for and around connection. It brings solace to some, terror to others. This paper, hence, sought not to provide an objective classification of this phenomenon, but rather a nuanced understanding of the multifaceted nature of the human mind in search of connection and the disconcerting direction taken by conglomerates attempting to capitalise on it.

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